

Hamza Khan

Senior Backend Engineer · Distributed Systems · Python · Kafka · AWS · GCP · Large-Scale Infrastructure
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SUMMARY

Backend engineer with deep experience designing and operating large-scale distributed systems in production. Built and maintained event-driven infrastructure processing 100K+ events/hour, serving 50M+ users with sub-second latency across globally distributed pipelines. Proven track record in system reliability, cloud infrastructure (AWS, GCP), and scalable backend architecture. Strong advocate of automated testing, CI-driven delivery, and measurable performance improvements.

EXPERIENCE

Senior Software Engineer · [Techolution](#)

Jun. 2025 – Present | Hyderabad, India

- Owned event-driven Kafka pipelines processing 100K+ real-time audio events per hour with sub-second end-to-end latency, supporting iHeartMedia's platform across 850+ radio stations and 50M+ monthly active users.
- Architected Python microservices across a distributed, multi-service infrastructure, delivered via trunk-based development and CI-driven small-batch releases for high availability.
- Designed Redis-backed distributed failure-tracking for LLM API workflows, reducing undetected task failures by 90% and on-call alerts by 30%.
- Shipped an AI-powered document ingestion pipeline (Python, Celery, Redis, multi-agent LLM) processing 10K+ documents/month, cutting manual processing time by 70%.
- Diagnosed and resolved async context-propagation bugs in real-time distributed pipelines, driving error rate from 15% to under 1%.
- Led database and backend workflow optimizations contributing to a 30% reduction in average response time, validated through unit and integration tests in CI.

Software Engineer · [Xempla, Enterprise Asset Management Platform](#)

Nov. 2021 – Jun. 2025 | Remote (Australia)

- Designed and scaled event-driven data pipelines for enterprise analytics workloads spanning 200+ client sites globally, handling both relational (PostgreSQL) and non-relational (MongoDB) data stores at scale.
- Re-architected core data infrastructure to reduce cloud compute load by 35%, saving an estimated \$80K/year — directly improving system efficiency and cost at scale.
- Maintained 99.7% uptime SLA across all managed distributed pipelines while resolving 30+ complex production incidents.
- Delivered RESTful API integrations and backend systems that accelerated a new product-line launch by 3 months and onboarded 15+ enterprise clients in year one.
- Expanded automated test coverage and introduced code-review standards, cutting post-deployment bug reports by 25%; mentored 3 junior engineers.

Backend Developer Intern · [WatchMyDC Analytics](#)

Mar. 2021 – Nov. 2021 | Remote (Finland)

- Implemented application monitoring across the software lifecycle to ensure overall system performance and reliability.
- Wrote scalable REST APIs in Python (Django) for a CRM platform serving 5K+ daily active users, and built internal features contributing to product success.
- Owned the full deployment cycle for the company's production website.
- Integrated a payment gateway into the CRM software, enabling end-to-end transaction handling.

PROJECTS

Kynstro | *Python, FastAPI, OpenAI, Redis, PostgreSQL, AWS*

kynstro.com

- Designed and shipped an end-to-end AI interview platform as the sole backend engineer, from distributed system design to cloud deployment on AWS.
- Built async LLM-powered interview pipelines, real-time scoring logic, and a multi-tenant API layer on FastAPI with full observability.
- Owned schema design, queueing infrastructure, LLM orchestration, and AWS deployment end-to-end.

TECHNICAL SKILLS

Languages: Python (primary), TypeScript, JavaScript

Distributed Systems: Apache Kafka, Event-Driven Architecture, Microservices, Async Programming, System Design

Cloud & Infrastructure: AWS (EC2, S3, SQS, Lambda), GCP, Docker, Kubernetes, GitHub Actions, Jenkins, CI/CD, Trunk-Based Development

Backend: FastAPI, Django, Flask, NestJS, REST APIs

Data & Messaging: Redis, Celery, PostgreSQL, MySQL, MongoDB

Testing & Quality: Unit Testing, Integration Testing, Code Review, Automated CI Pipelines

AI / LLM: OpenAI APIs, AI Agents, LLM Integrations, Multi-Agent Pipelines

EDUCATION

Integral University — Lucknow, India

Bachelor in Computer Application · CGPA: 8.9/10

2020 – 2023